



Build a RAG system - From Beginners to Advanced

Stephen Batifol | Zilliz

Tirana Tech Meetup, July 4th



Speaker



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✕ [@stephenbt1](https://twitter.com/stephenbt1)





Easy Setup

`pip install
pymilvus` to start
coding in a notebook
within seconds.



Reusable Code

Write once, and
deploy with one line
of code into the
production
environment



Integration

Plug into OpenAI,
Langchain,
LlamaIndex, and
many more



Feature-rich

Dense & sparse
embeddings,
filtering, reranking
and beyond



Milvus is an open-source vector database for **GenAI** projects. `pip install` on your laptop, plug into popular AI dev tools, and push to production with a single line of code.



27K+

GitHub
Stars



2,600

Forks



25M+

Downloads



250+

Contributors

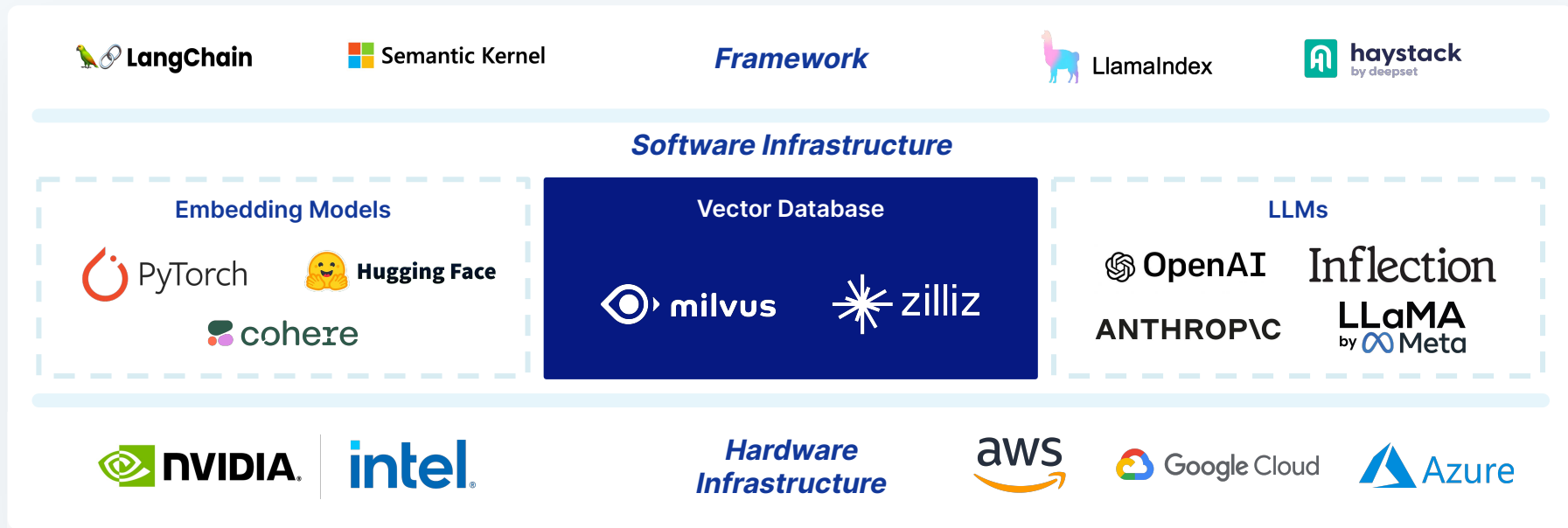
Seamless integration with all popular AI toolkits



VOYAGE AI



Well-connected in LLM infrastructure to enable RAG use cases



Build Once Deploy Anywhere

| GenAI Application

Milvus Client Code

```
1 client = MilvusClient(...)
2 client.insert(...)
3 client.search(...)
```

| Milvus Lite



| Milvus

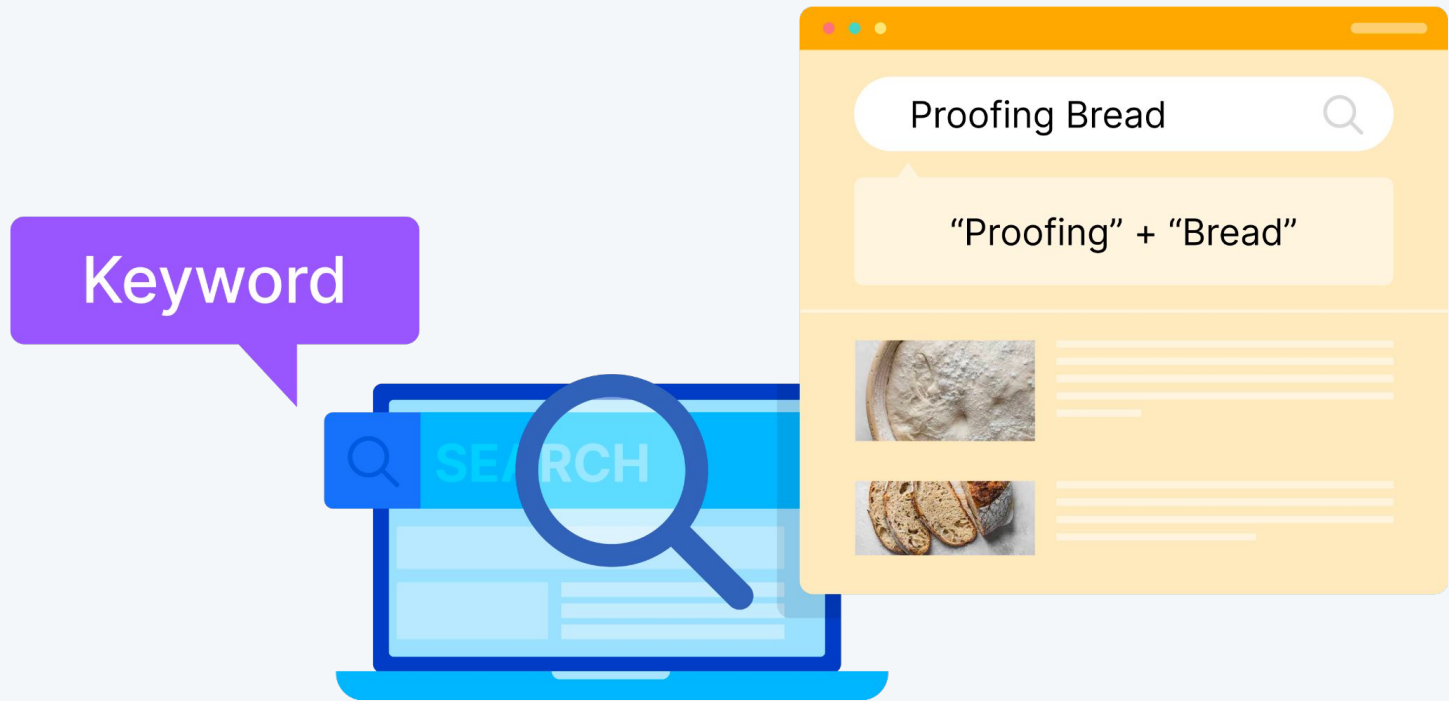


01

Introduction to Vector DB and Vector Search



Traditional database was built upon exact search



...which misses context, semantic meaning, and user intent

Q | Apple



VS.



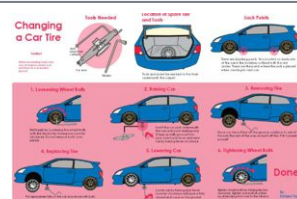
Q | Rising dough

Rising Dough ✓

VS.

Proofing Bread ✗

Q | Change car tire

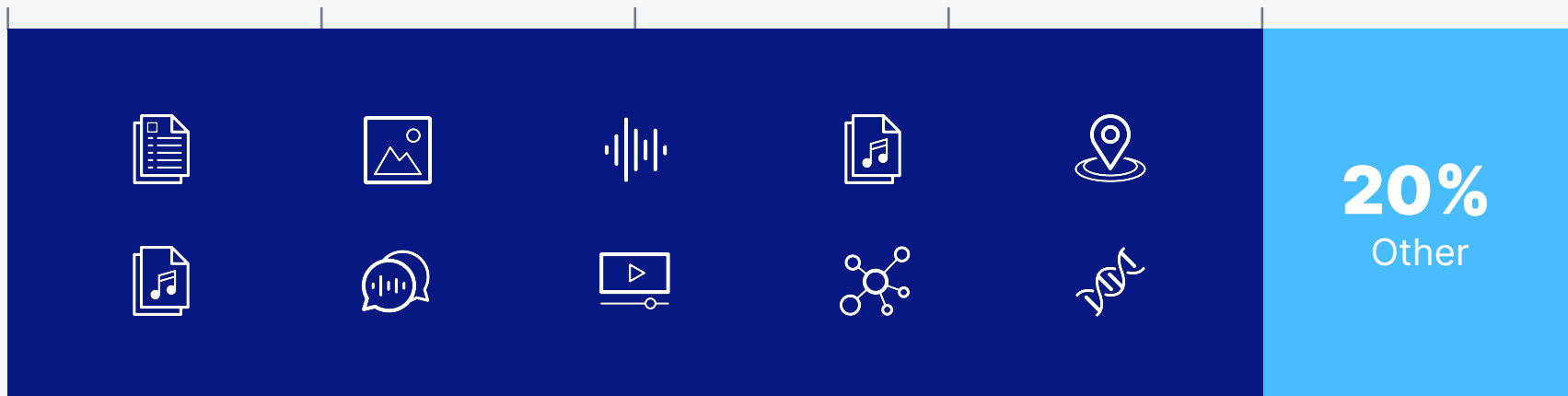


VS.



...and cannot process increasingly growing unstructured data

◀ **80%** newly generated data in 2025
will be unstructured data ▶



Common AI Use Cases



Retrieval Augmented Generation (RAG)

Expand LLMs' knowledge by incorporating external data sources into LLMs and your AI applications.



Recommender System

Match user behavior or content features with other similar ones to make effective recommendations.



Text/ Semantic Search

Search for semantically similar texts across vast amounts of natural language documents.



Image Similarity Search

Identify and search for visually similar images or objects from a vast collection of image libraries.



Video Similarity Search

Search for similar videos, scenes, or objects from extensive collections of video libraries.



Audio Similarity Search

Find similar audios in large datasets for tasks like genre classification or speech recognition



Molecular Similarity Search

Search for similar substructures, superstructures, and other structures for a specific molecule.



Anomaly Detection

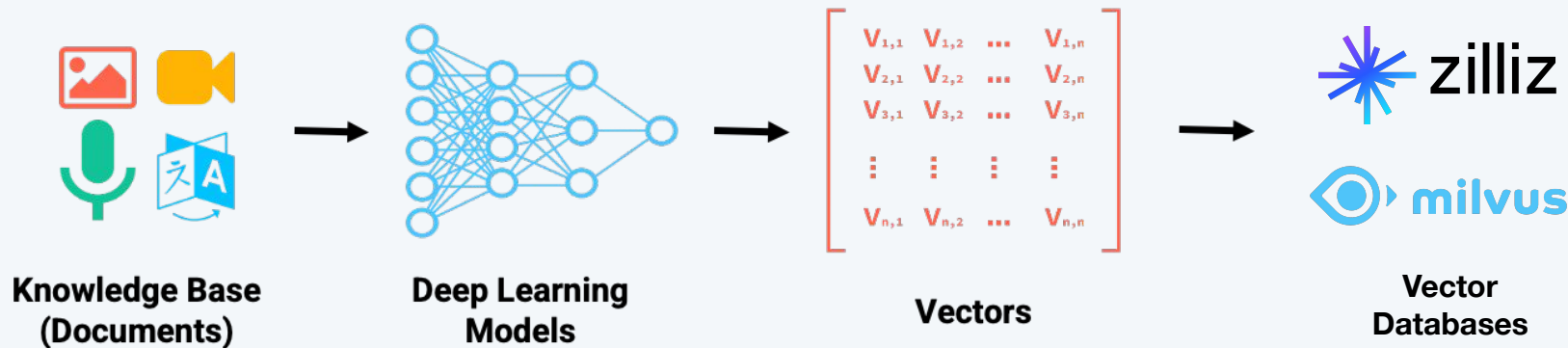
Detect data points, events, and observations that deviate significantly from the usual pattern



Multimodal Similarity Search

Search over multiple types of data simultaneously, e.g. text and images

Where do Vectors Come From?

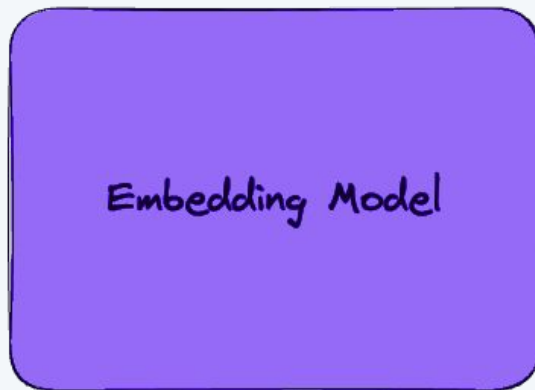


Embeddings Models



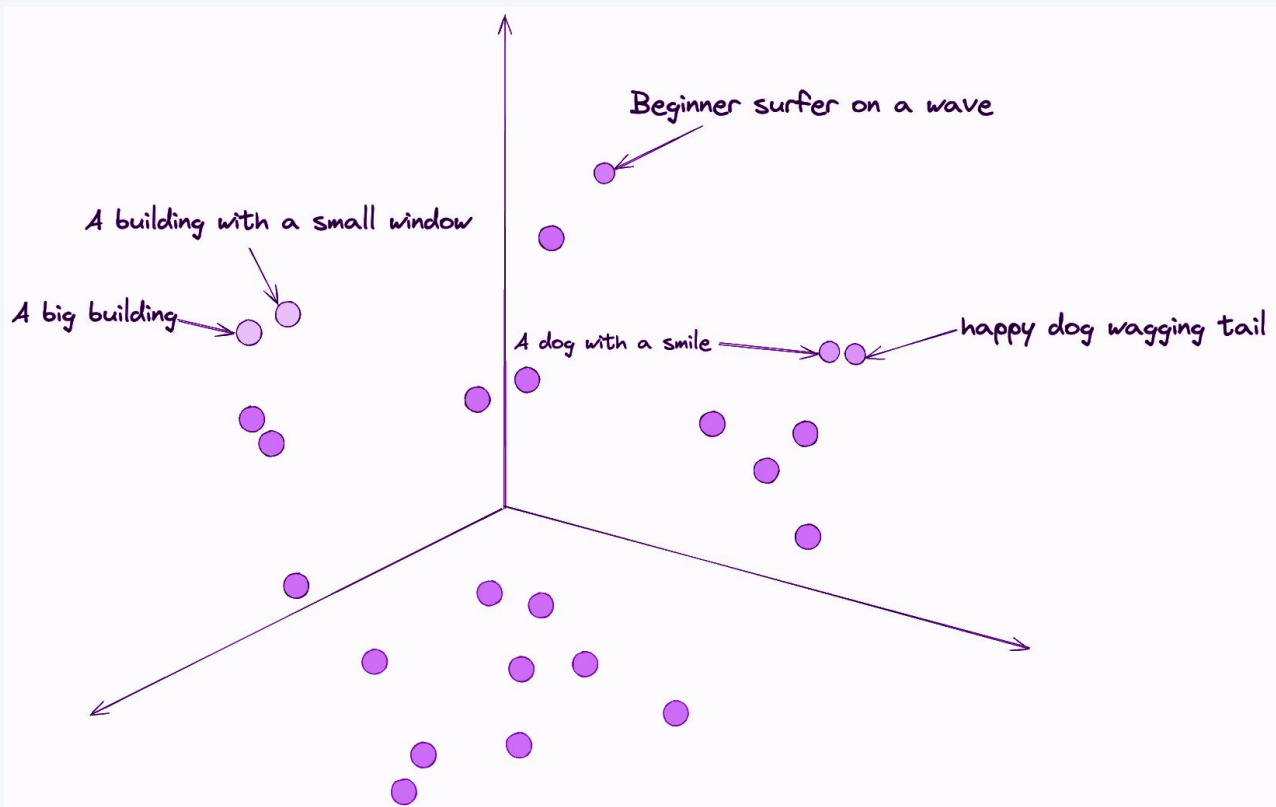
Vector Embedding

happy dog wagging tail



$[-0.096, -0.026, -0.044, 0.012, \dots, -0.011]$

Vector Space

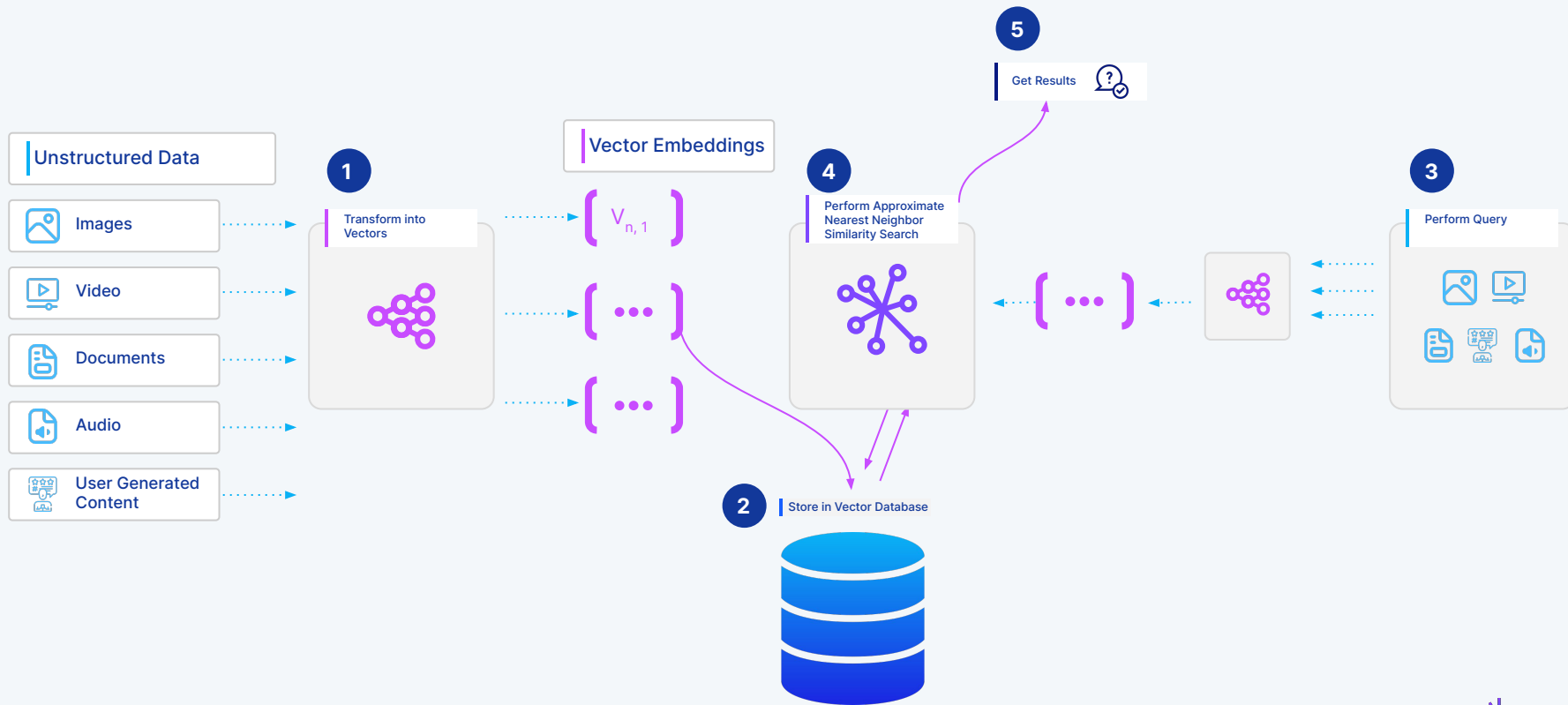


02

How do Vector Databases Work?



How Similarity Search Works



Example Entry

```
"id": "https://towardsdatascience.com/detection-of-credit-card-fraud-with-an-autoencoder-9275854  
"embedding": [-0.042092223,-0.0154002765,-0.014588429,-0.031147376,0.03801204,0.013369046,0  
"date": "2023-06-01"  
"paragraph": "We define an anomaly as follows:"  
"reading_time": "11"  
"subtitle": "A guide for the implementation of an anomaly..."  
"publication": "Towards Data Science"  
"responses": "1"  
"article_url": "https://towardsdatascience.com/detection-of-credit-card-fraud-with-an-autoencoder-  
"title": "Detection of Credit Card Fraud with an Autoencoder"  
"claps": "229"
```

↑ Hide 6 fields

🔍 Vector search

03

Indexes



Indexing strategies

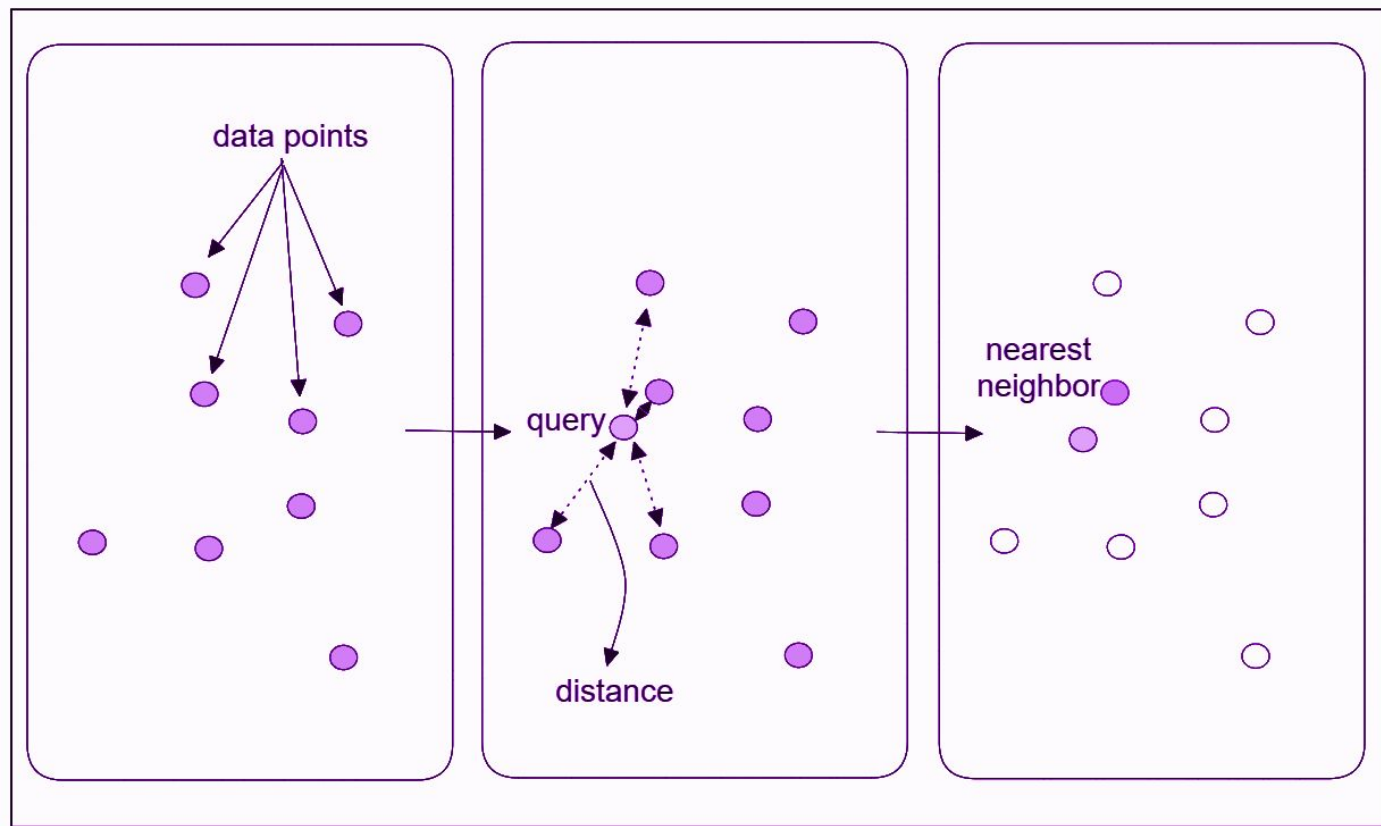
- Tree based
- Graph based
- Hash based
- Cluster based

Category	Index	Accuracy	Latency	Throughput	Index Time	Cost
Graph-based	Cagra (GPU)	High	Low	Very High	Fast	Very High
	HNSW	High	Low	High	Slow	High
	DiskANN	High	High	Mid	Very Slow	Low
Quantization-based or cluster-based	ScaNN	Mid	Mid	High	Mid	Mid
	IVF_FLAT	Mid	Mid	Low	Fast	Mid
	IVF + Quantization	Low	Mid	Mid	Mid	Low



FLAT

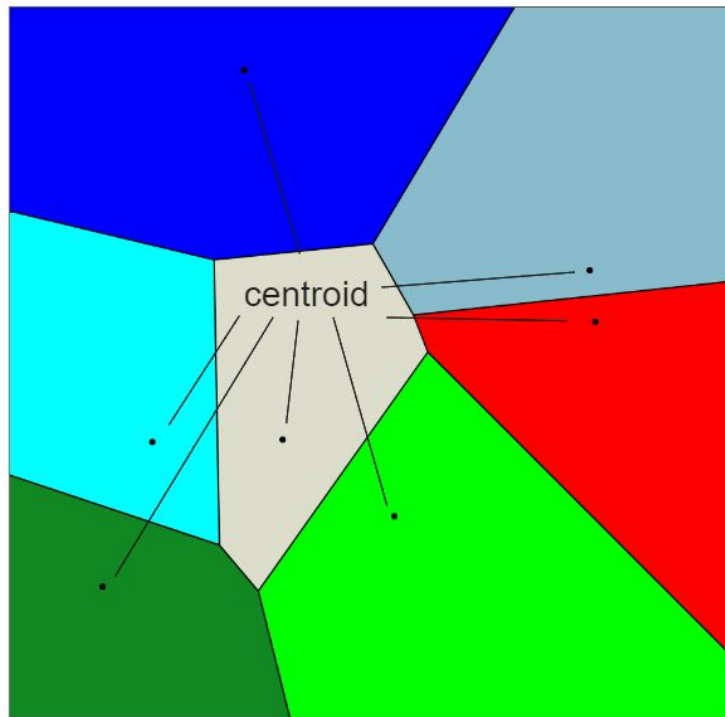
FLAT Index



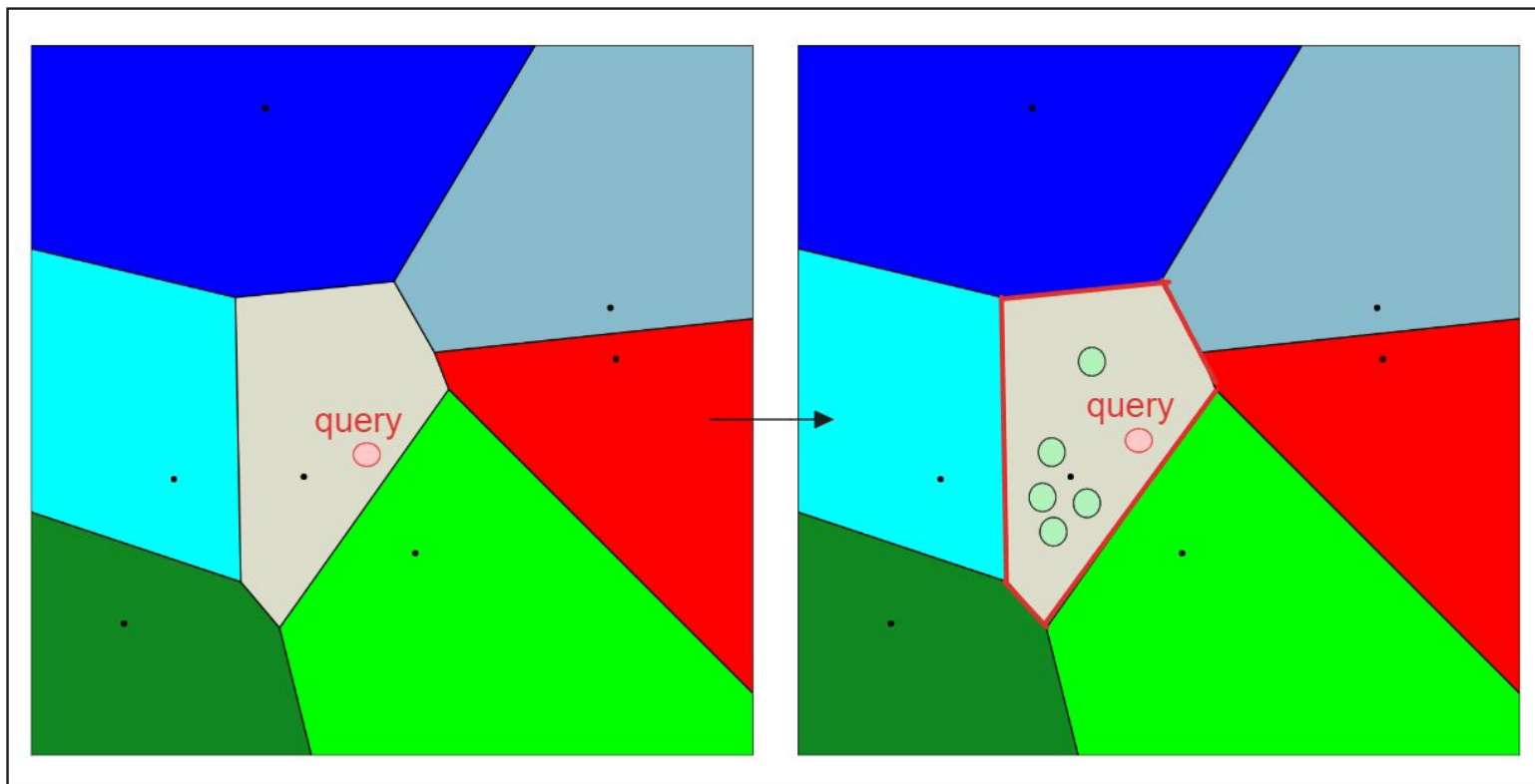
Inverted File FLAT

(IVF-FLAT)

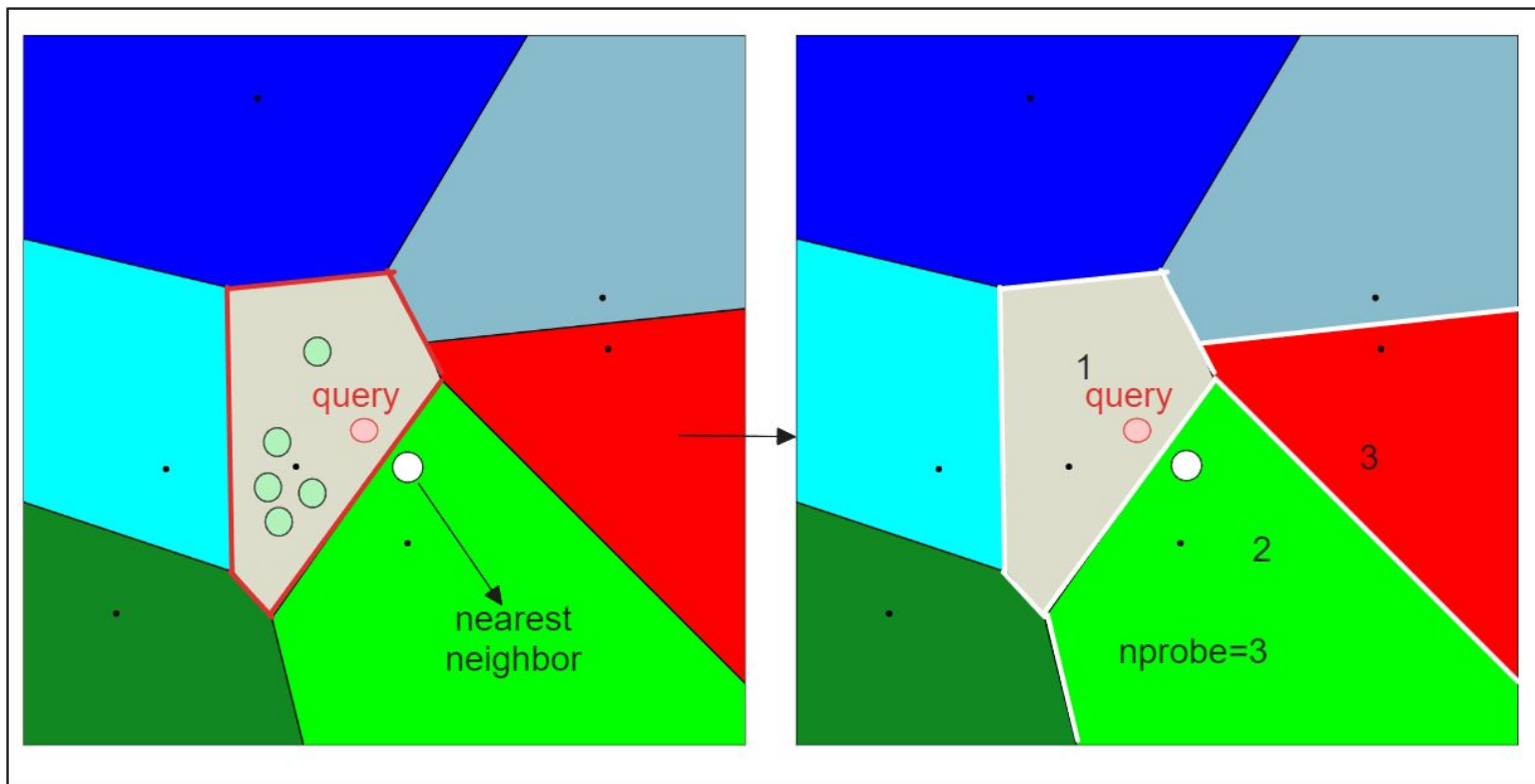
IVF-FLAT Index



IVF-FLAT Index



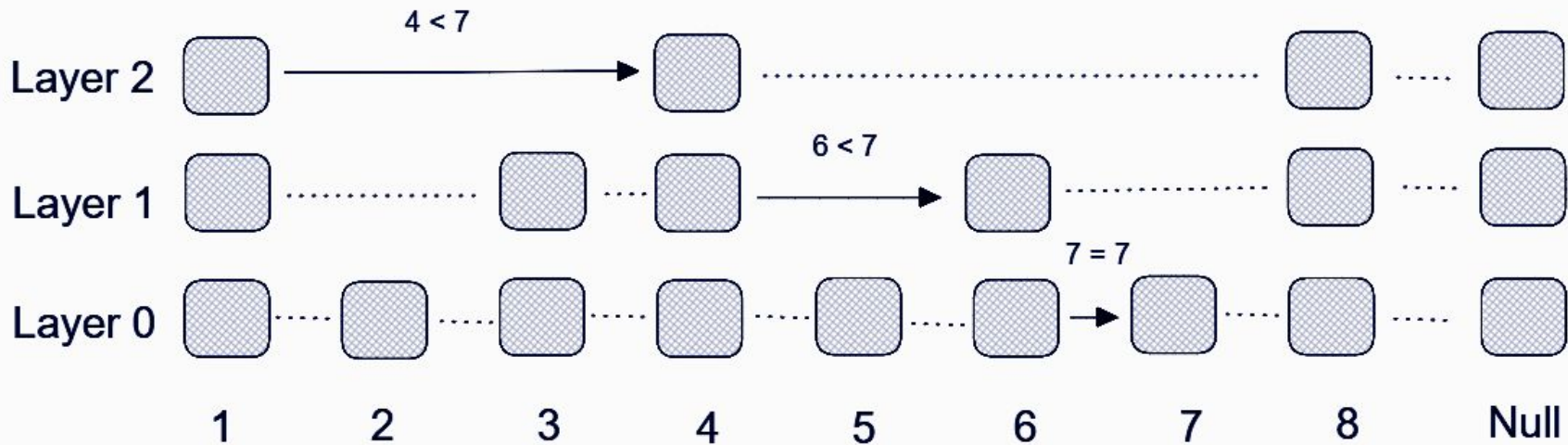
IVF-FLAT Index



Hierarchical Navigable Small World (HNSW)

HNSW - Skip List

Finding element 7



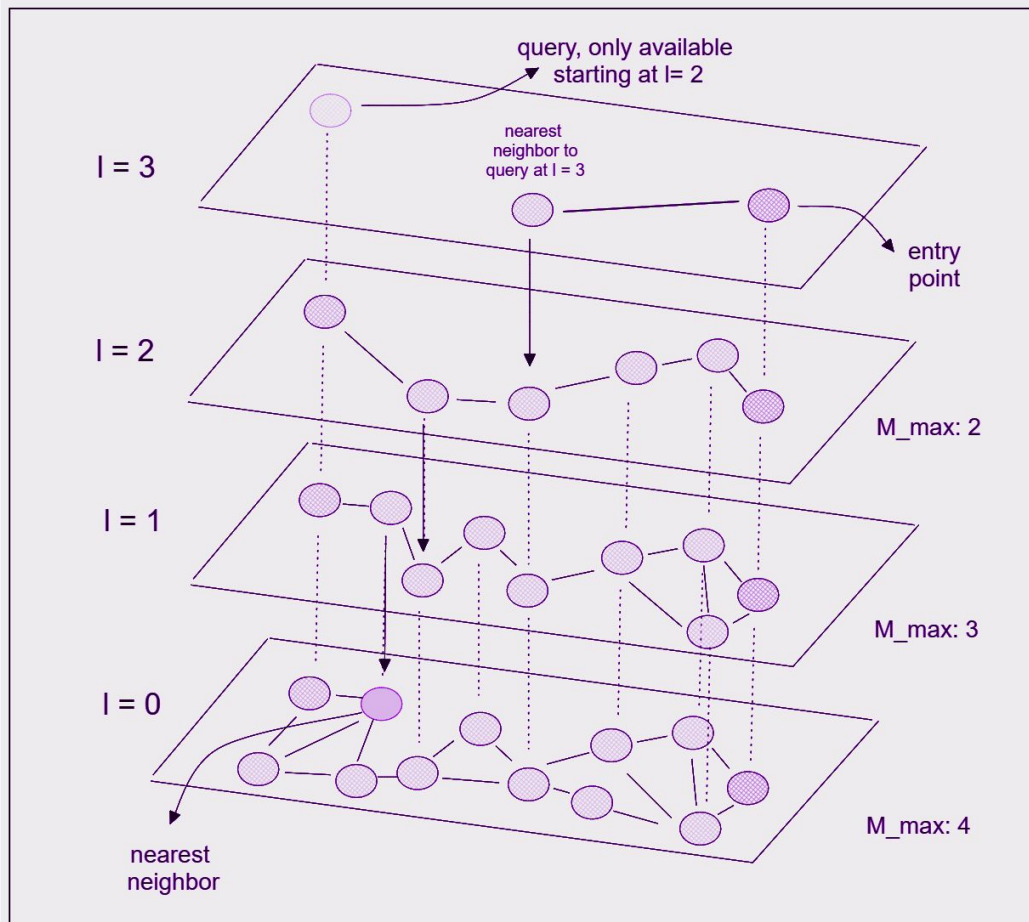
HNSW - NSW Graph

- Built by randomly shuffling data points and inserting them one by one, with each point connected to a predefined number of edges (M).

⇒ Creates a graph structure that exhibits the "small world".

⇒ Any two points are connected through a relatively short path.

HNSW



Category	Index	Accuracy	Latency	Throughput	Index Time	Cost
Graph-based	Cagra (GPU)	High	Low	Very High	Fast	Very High
	HNSW	High	Low	High	Slow	High
	DiskANN	High	High	Mid	Very Slow	Low
Quantization-based or cluster-based	ScaNN	Mid	Mid	High	Mid	Mid
	IVF_FLAT	Mid	Mid	Low	Fast	Mid
	IVF + Quantization	Low	Mid	Mid	Mid	Low

Picking an Index

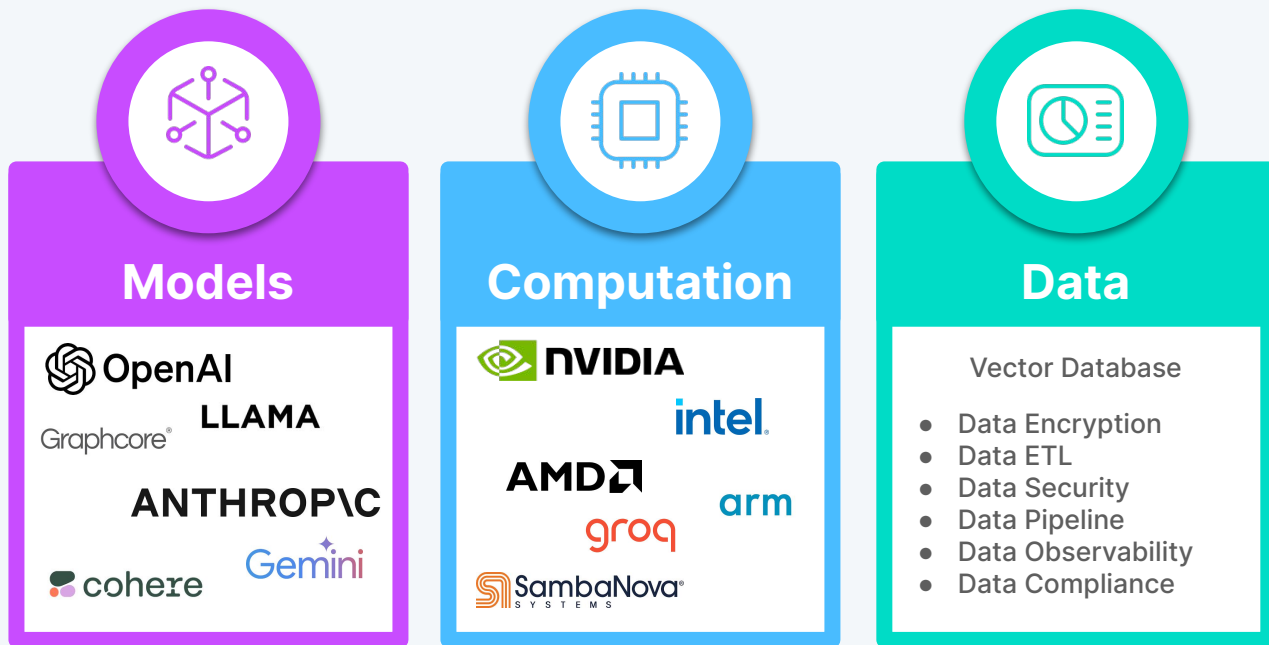
- 100% Recall – Use **FLAT** search if you need 100% accuracy
- $10\text{MB} < \text{index_size} < 2\text{GB}$ – Standard IVF
- $2\text{GB} < \text{index_size} < 20\text{GB}$ – Consider **PQ** and **HNSW**
- $20\text{GB} < \text{index_size} < 200\text{GB}$ – Composite Index, **IVF_PQ** or **HNSW_SQ**
- Disk-based indexes

The GenAI Boom





Three Pillars of GenAI & the opportunities they bring



RAG

(Retrieval Augmented Generation)



Basic Idea

Use RAG to **force** the **LLM** to **work with your data**
by injecting it via a vector database like **Milvus**

Why RAG?

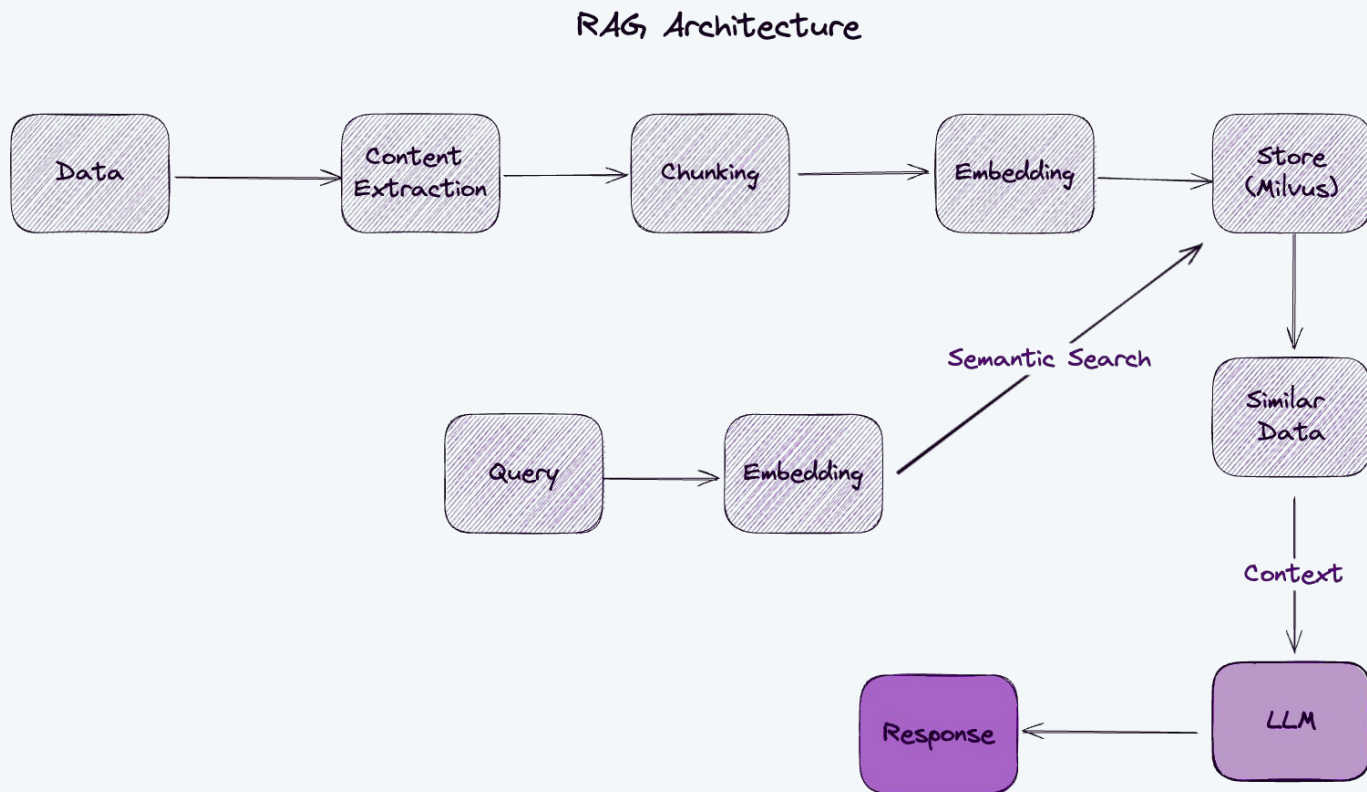
RAG vs. LLM

- Knowledge of LLM is out-of-date
- LLM can not get your private knowledge
- Help reduce Hallucinations
- Transparency and interpretability

RAG vs. Fine-tune

- Fine-tune is expensive
- Fine-tune spent much time
- RAG is pluggable

Basic RAG Architecture



5 lines starter

```
1 # load and parse
2 documents = SimpleDirectoryReader("data").load_data()
3
4 # embed and index
5 index = VectorStoreIndex.from_documents(documents)
6
7 # generate query engine
8 query_engine = index.as_query_engine()
9
10 # run queries
11 response = query_engine.query("What did the author do growing
12 up?")
```

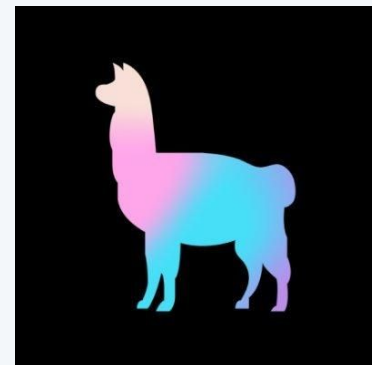
04

Basic RAG with Milvus



Llama-Index

- Framework for building LLM Applications
- Focus on retrieving data and integrating with LLMs
 - Loading the Data
 - Chunk & Chunk Overlap
- Integrations with most AI popular tools



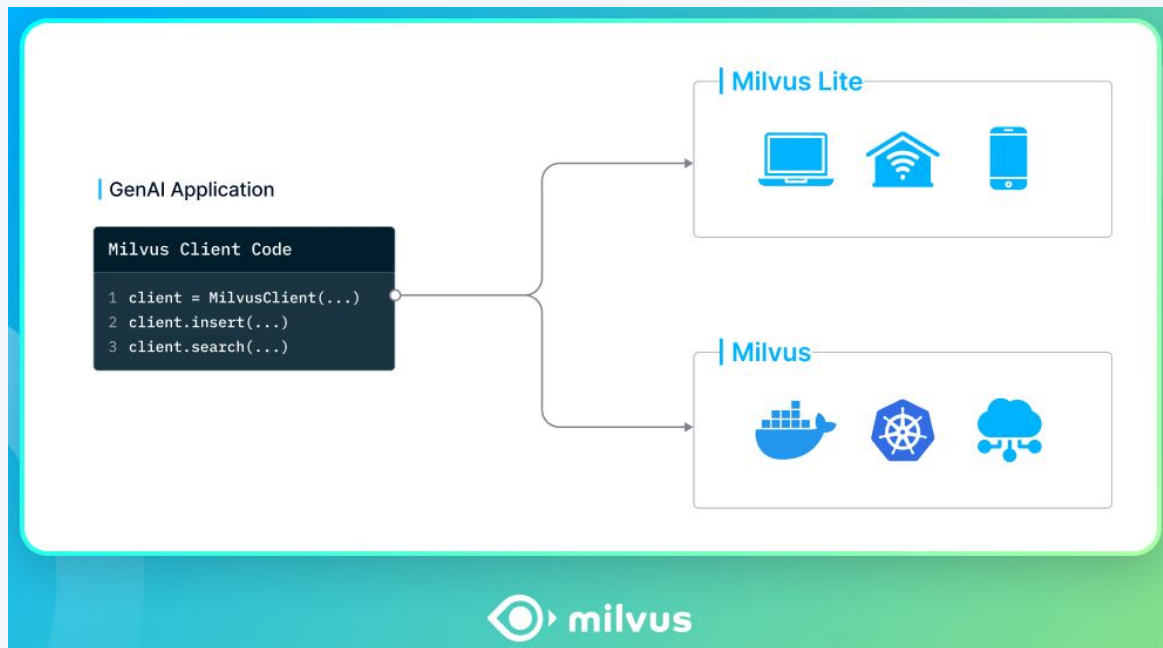
Ollama

- Run LLMs Locally
- Embeddings Models



Milvus Lite

```
pip install pymilvus
```



05

Embeddings



Embeddings Models



Examining Embeddings

Picking a model

What to embed

Metadata

Embeddings Strategies

Level 1: Embedding Chunks Directly

Level 2: Embedding Sub and Super Chunks

Level 3: Incorporating Chunking and Non-Chunking Metadata

Metadata Examples

Chunking

- Paragraph position
- Section header
- Larger paragraph
- Sentence Number
- ...

Non-Chunking

- Author
- Publisher
- Organization
- Role Based Access Control
- ...

Takeaway:

Your embeddings strategy depends on your **accuracy**,
cost, and **use case** needs

06

Chunking



Chunking Considerations

Chunk Size

Chunk Overlap

Character Splitters

Chunk Size=50, Overlap=0

Text is: Table of Contents

UNITED STATES

SECURITIES

Text is: AND EXCHANGE COMMISSION

Washington, D.C.

Text is: 20549

FORM 10-Q

(Mark

Text is: One)

☒ QUARTERLY REPORT

Text is: PURSUANT TO SECTION 13 OR

Text is: 15(d) OF THE SECURITIES EXCHANGE

Text is: ACT OF 1934

For the quarterly period ended

Text is: June 30,

Text is: 2023

OR

☐ TRANSITION REPORT

Text is: PURSUANT TO SECTION 13 OR

Chunk Size=128, Overlap=20

```
Text is: Table of Contents
UNITED STATES
SECURITIES AND EXCHANGE COMMISSION
Washington, D.C. 20549
FORM 10-Q
(Mark One)
☒ QUARTERLY REPORT PURSUANT TO SECTION 13 OR 15(d) OF THE SECURITIES EXCHANGE ACT OF 1934
For the quarterly period ended June 30, 2023
OR
☐ TRANSITION REPORT

Text is: the quarterly period ended June 30, 2023
OR
☐ TRANSITION REPORT PURSUANT TO SECTION 13 OR 15(d) OF THE SECURITIES EXCHANGE ACT OF 1934
For the transition period from _____ to _____
Commission file number 001-39419
WEWORK INC.
(Exact name of registrant as specified in its charter)
Delaware

Text is: INC.
(Exact name of registrant as specified in its charter)
Delaware 85-1144904
(State or other jurisdiction of incorporation or
organization)(I.R.S. Employer Identification No.)
12 East 49th Street, 3rd floor
New York, NY
(Address of principal executive offices)10017
(zip code)
(646) 389-3922
Registrant's telephone number, including area

Text is: code)
(646) 389-3922
Registrant's telephone number, including area code
Securities registered pursuant to Section 12(b) of the Act:
Title of each class Trading Symbol(s) Name of each exchange on which registered
Class A common stock, $0.0001 per share WE The New York Stock Exchange
Redeemable warrants, exercisable for shares of Class A common stock at
an exercise
```

Chunk Size=256, Overlap=50

Text is: Table of Contents
UNITED STATES
SECURITIES AND EXCHANGE COMMISSION
Washington, D.C. 20549
FORM 10-Q
(Mark One)
☒ QUARTERLY REPORT PURSUANT TO SECTION 13 OR 15(d) OF THE SECURITIES EXCHANGE ACT OF 1934
For the quarterly period ended June 30, 2023
OR
☐ TRANSITION REPORT PURSUANT TO SECTION 13 OR 15(d) OF THE SECURITIES EXCHANGE ACT OF 1934
For the transition period from _____ to _____
Commission file number 001-39419
WEWORK INC.
(Exact name of registrant as specified in its charter)
Delaware 85-1144904
(State or other jurisdiction of incorporation or organization)(I.R.S. Employer Identification No.)
12 East 49th Street, 3rd floor
New York, NY
(Address of principal executive offices)10017
(zip code)
(646)

Text is: or other jurisdiction of incorporation or organization)(I.R.S. Employer Identification No.)
12 East 49th Street, 3rd floor
New York, NY
(Address of principal executive offices)10017
(zip code)
(646) 389-3922
Registrant's telephone number, including area code
Securities registered pursuant to Section 12(b) of the Act:
Title of each class Trading Symbol(s) Name of each exchange on which registered
Class A common stock, \$0.0001 per share WE The New York Stock Exchange
Redeemable warrants, exercisable for shares of Class A common stock at an exercise price of \$11.50 per share WE WS The New York Stock Exchange
Class A Common Stock Purchase Rights - The New York Stock Exchange
Securities registered pursuant to section 12(g) of the Act:
None

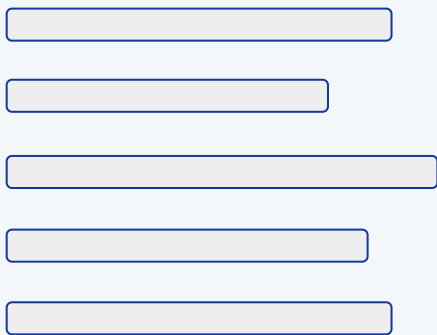
Indicate by check mark whether the registrant: (1) has filed all reports required to be filed by Section 13 or 15(d) of the Securities Exchange Act of 1934

SemanticChunker

Text chunk: ['Table of Contents\nUNITED STATES\nSECURITIES AND EXCHANGE COMMISSION\nWashington, D.C. 20549\nFORM 10-Q\n(Mark One)\nQUARTERLY REPORT PURSUANT TO SECTION 13 OR 15(d) OF THE SECURITIES E XCHANGE ACT OF 1934\nFor the quarterly period ended June 30, 2023\nOR\nTRANSITION REPORT PURSUANT TO SECTION 13 OR 15(d) OF THE SECURITIES EXCHANGE ACT OF 1934\nFor the transition period from _____ to _____\nCommission file number 001-39419\nNEWWORK INC.', '(Exact name of registrant as specified in its charter)\nDelaware 85-1144904\n(State or other jurisdiction of incorporation or\norganization)\n(I.R.S. Employer Identification No.)\n12 East 49th Street, 3rd floor\nNew York, NY\n(Address of principal executive offices)\n10017\n(zip code)\n(646) 389-3922\nRegistrant's telephone number, including area code\nSecurities registered pursuant to Section 12(b) of the Act:\nTitle of each class Trading Symbol(s) Name of each exchange on which registered\nClass A common stock, \$0.0001 per share WE The New York Stock Exchange\nRedeemable warrants, "exercisable for shares of Class A common stock at\nan exercise price of \$11.50 per share WE WS The New York Stock Exchange\nClass A Common Stock Purchase Rights - The New York Stock Exchange\nSecurities registered pursuant to section 12(g) of the Act:\nNone\nIndicate by check mark whether the registrant: (1) has filed all reports required to be filed by Section 13 or 15(d) of the Securities Exchange Act of 1934 during the preceding 12 months (or for such shorter period that the registrant was required to file\nsuch reports); and (2) has been subject to such filing requirements for the past 90 days.', 'Yes x No o\nIndicate by check mark whether the registrant has submitted electronically and posted on its corporate web site, if any, every Interactive Data File required to be submitted and posted pursuant to Rule 405 of Regulation S-T (\$232.405 of this chapter)\nduring the preceding 12 months (or for such shorter period that the registrant was required to submit and post such files). Yes x No o\nIndicate by check mark whether the registrant is a large accelerated filer, an accelerated filer, a non-accelerated filer, or a smaller reporting company.', 'See the definitions of "large accelerated filer," "accelerated filer" and "smaller reporting company"\nin Rule 12b-2 of the Exchange Act. (Check one):\nLarge accelerated filer o Accelerated filer x\nNon-accelerated filer o Smaller reporting company o\nEmerging growth company o\nIf an emerging growth company, indicate by check mark if the registrant has elected not to use the extended transition period for complying with any new or revised financial accounting standards provided pursuant to Section 13(a) of the Exchange Act.', "o\nIndicate by check mark whether the registrant is a shell company (as defined in Rule 12b-2 of the Act). Yes o No x\nIndicate the number of shares outstanding of each of the issuer's classes of common stock, as of the latest practicable date: As of August 7, 2023, there were 2,110,280,756 shares of Class A common stock, par value \$0.0001 per share, and 19,938,089\nshares of Class C common stock, par value \$0.0001 per share, issued and outstanding."]

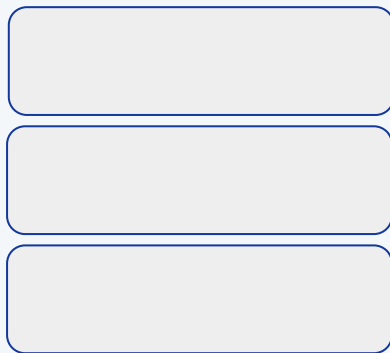
Text chunk: ['Table of Contents\nNewWork Inc.\nFORM 10-Q\nTHREE AND SIX MONTHS ENDED JUNE 30, 2023\nTABLE OF CONTENTS\nPage\nPart I - Financial Information\nCautionary Note Regarding Forward-Looking Statements 3\nItem 1.', 'Financial Statements and Supplementary Data 6\nCondensed Consolidated Balance Sheets 7\nCondensed Consolidated Statements of Operations 8\nCondensed Consolidated Statements of Comprehensive Loss 9\nCondensed Consolidated Statement of Changes in Convertible Preferred Stock, Noncontrolling Interests, and Equity 10\nCondensed Consolidated Statements of Cash Flows 12\nNotes to the Condensed Consolidated Financial Statements 13\nItem 2. Management's Discussion and Analysis of Financial Condition and Results of Operations 70\nItem 3. Quantitative and Qualitative Disclosures About Market Risk 113\nItem 4. Controls and Procedures 114\nPart II - Other Information\nItem 1.', 'Legal Proceedings 115\nItem 1A. Risk Factors 116\nItem 1B. Unresolved Staff Comments 116\nItem 2. Unregistered Sales of Equity Securities and Use of Proceeds 116\nItem 3. Defaults Upon Senior Securities 117\nItem 4. Mine Safety Disclosures 117\nItem 5. Other Information 117\nItem 6. Exhibits 119\nSignature s 119']

How Does Your Data Look?



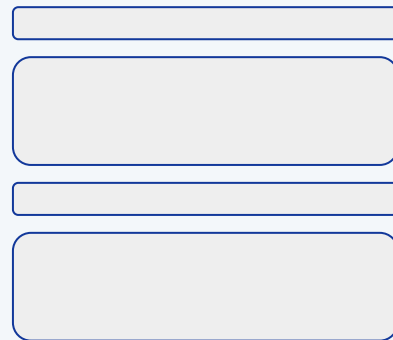
Conversation

Data



Documentation

Data



Lecture or Q/A

Data

Takeaway:

Your chunking strategy depends on **what your data looks like** and **what you need from it**.



Demo!

Naive RAG is limited



Jo Kristian Bergum ✓ @jobergum · 23h

...

It's rare to see a sensible HN comment news.ycombinator.com/item?id=404435... @jxnlco

reply

▲ Xenoamorphous 18 minutes ago | prev | next [-]

Why is there so much buzz about RAG?

Isn't it basically a traditional search (either keyword based, vector based -embeddings have been around for years-, or a combination of both) where you take the top N results (usually not even full docs, but chunks due to context length limitations) and pass them to an LLM to regurgitate a response (hopefully without hallucinations), instead of simply listing the results right away? I think some implementations also ask the LLM to rewrite the user query to "capture the user intent".

What I'm missing here? What makes it so useful?

reply

💬 22

↻ 34

❤ 262

📊 67K

🔖 ⬆

Naive RAG failure mode

Summarization

Naive RAG failure mode

Implicit data

Naive RAG failure mode

Multi-part questions

RAG is necessary but not sufficient



Good dishes come from good ingredients

- Data collection
- Data cleaning
- Parsing & Chunking

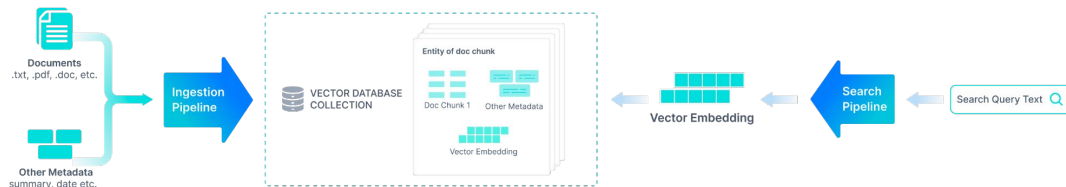


Pipelines

Easy. High-quality. Scalable.

Simplify and streamline the conversion of unstructured data into state-of-the-art vector embeddings, using intuitive UI and Restful APIs.

Zilliz Cloud Pipelines

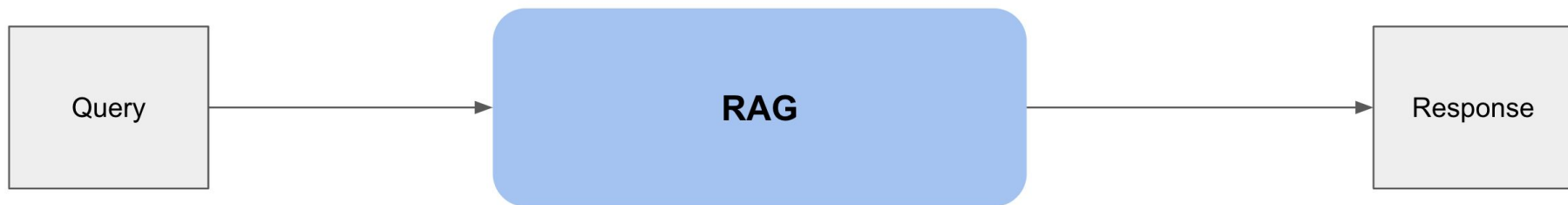


Simplify the workflow for developers, from converting unstructured data into searchable vectors to retrieving them from vector databases

Deliver excellence in every phase of vector search pipeline development and deployment, regardless of their expertise

Ensure scalability for managing large datasets and high-throughput queries, maintaining high performance with min. customization or infra changes

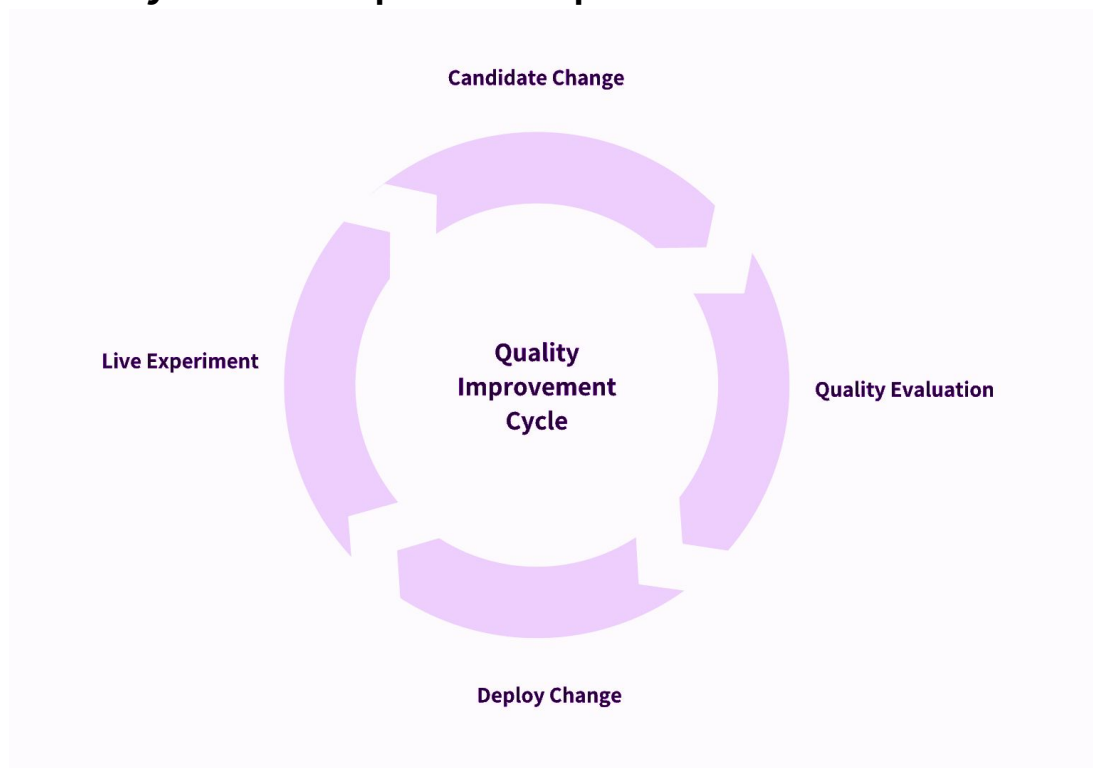
Naive RAG Pipeline



- ⚠ Single-shot
- ⚠ No query understanding/planning
- ⚠ No tool use
- ⚠ No reflection, error correction
- ⚠ No memory (stateless)

First thing first

Measure it before you attempts to improve it!



07

Advanced RAG techniques



Types of RAG Enhancement Techniques

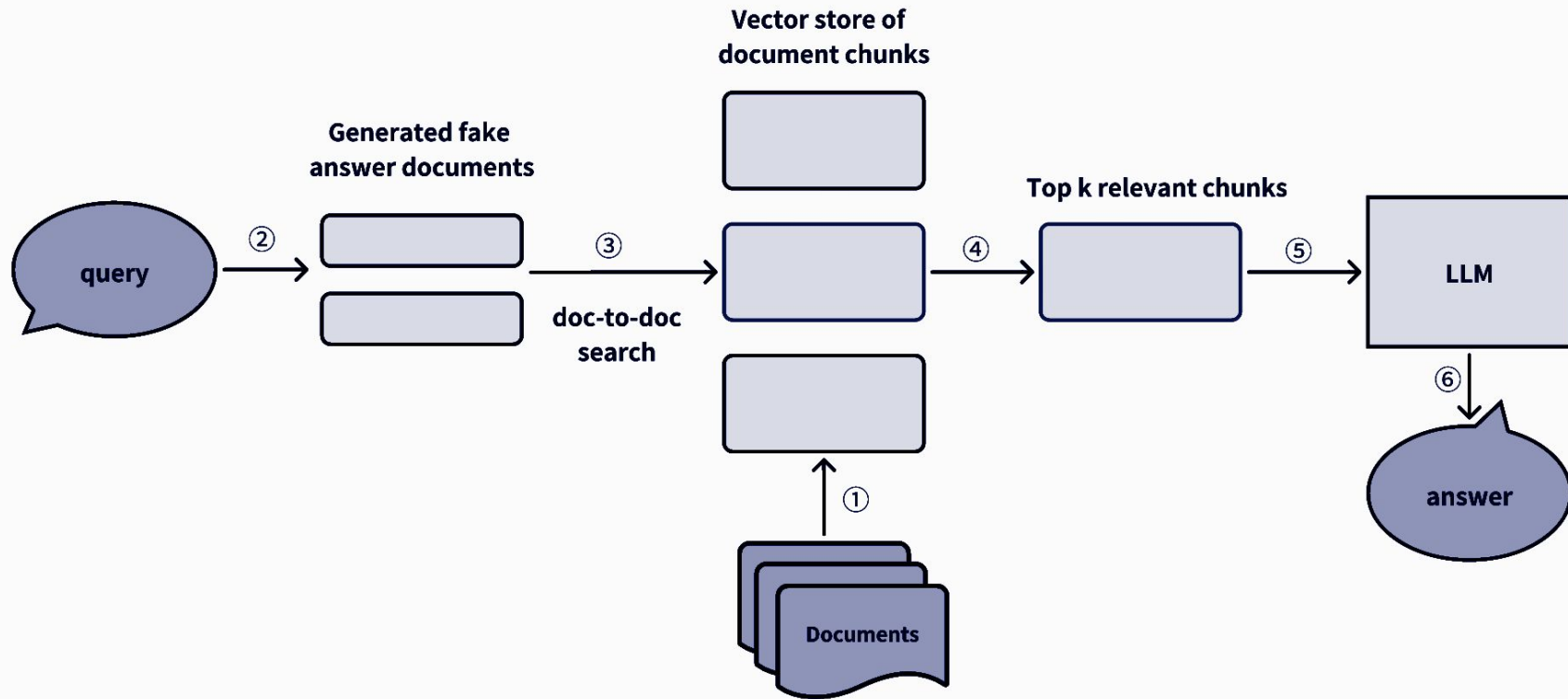
- **Divide & Conquer**
 - **Query Enhancement:** better express or process the query intent.
 - **Indexing Enhancement:** data cleanup, better parser and chunking
 - **Retriever Enhancement:** more retrievers and hybrid search strategy
- **Thinking outside the box**
 - Agents? Other tools than retriever?

a)

Query Enhancement



HyDE

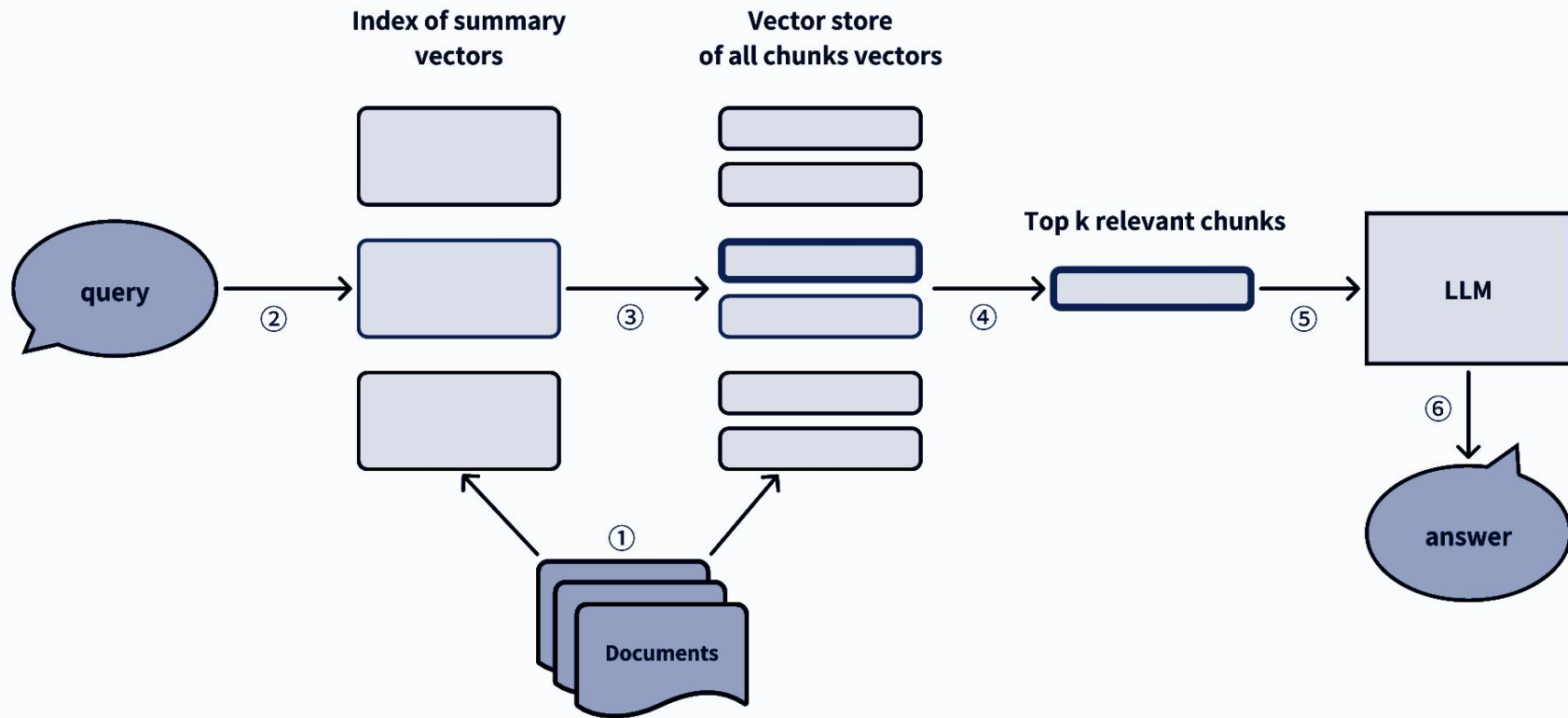


b)

Indexing Enhancement



Hierarchical index

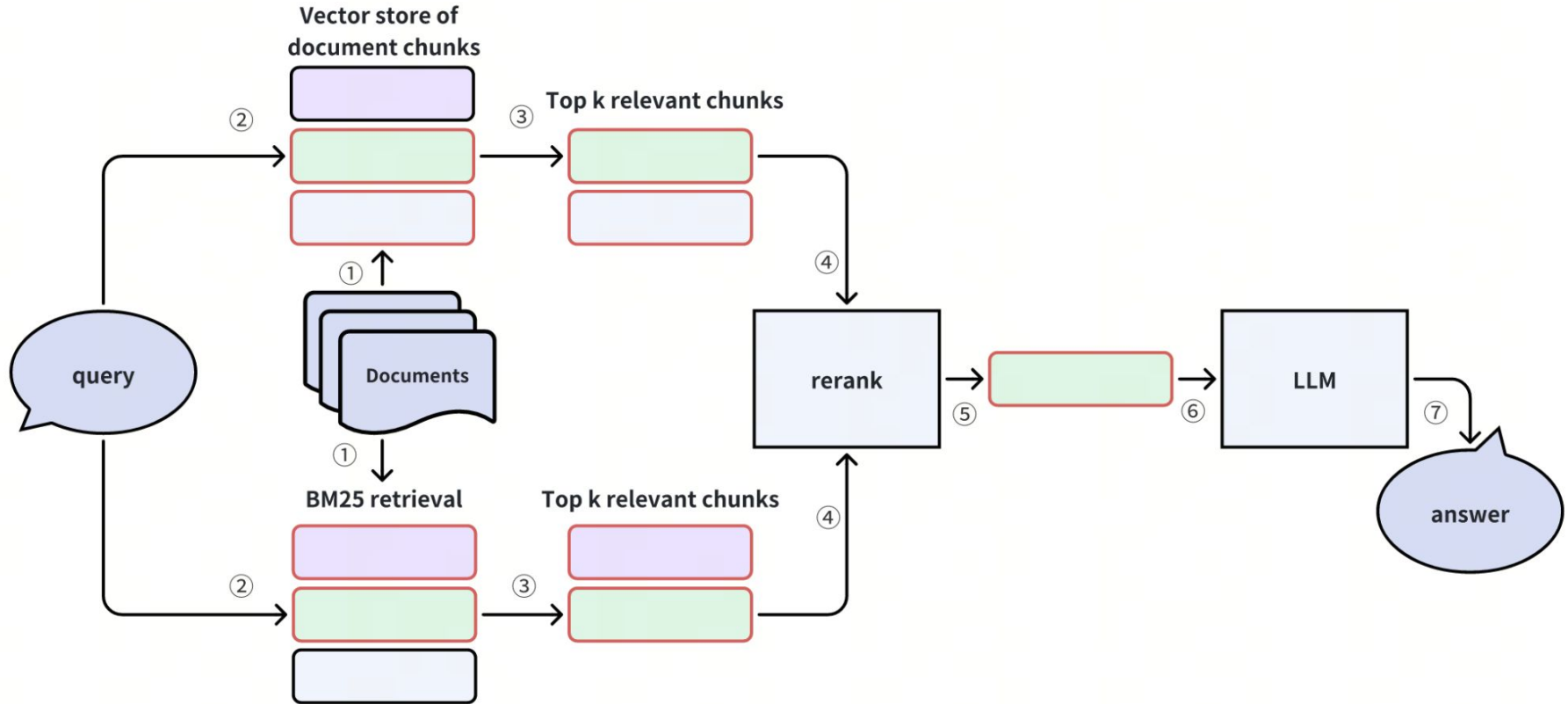


c)

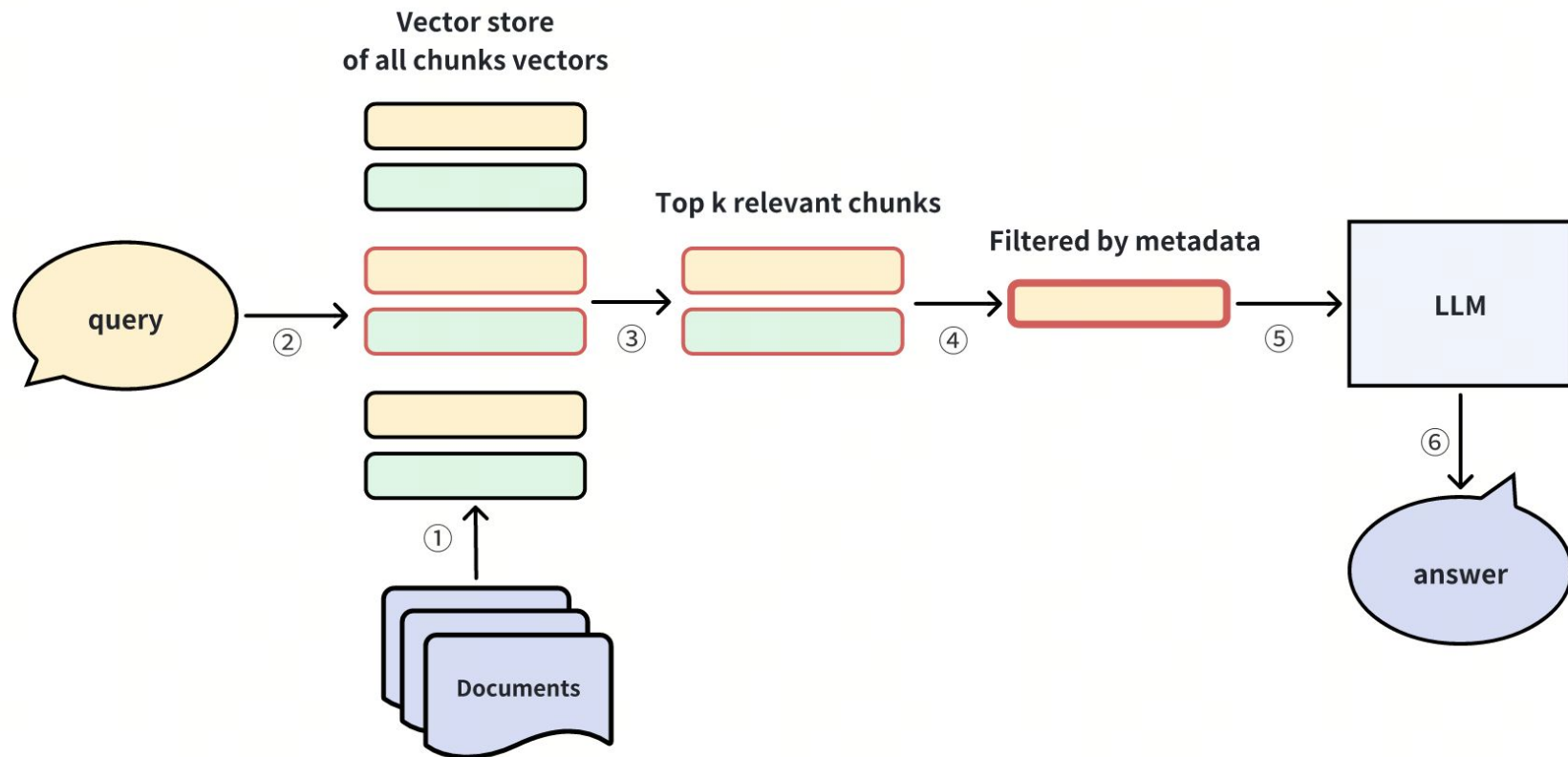
Retriever Enhancement



Hybrid Retrieve & reranking



Meta-data Filtering

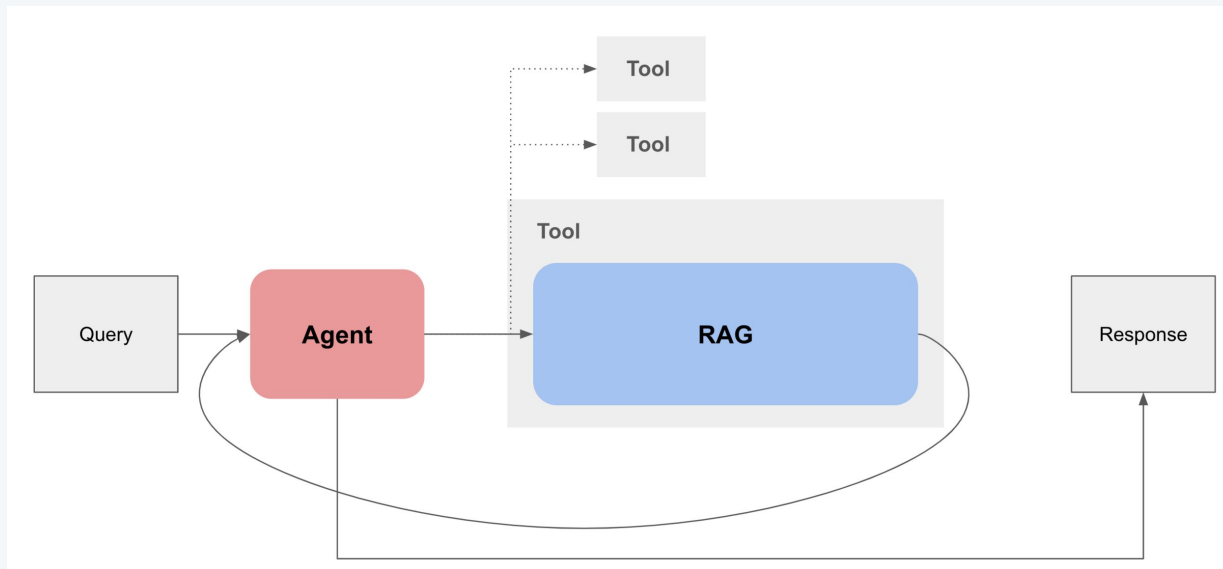


08

Agentic RAG

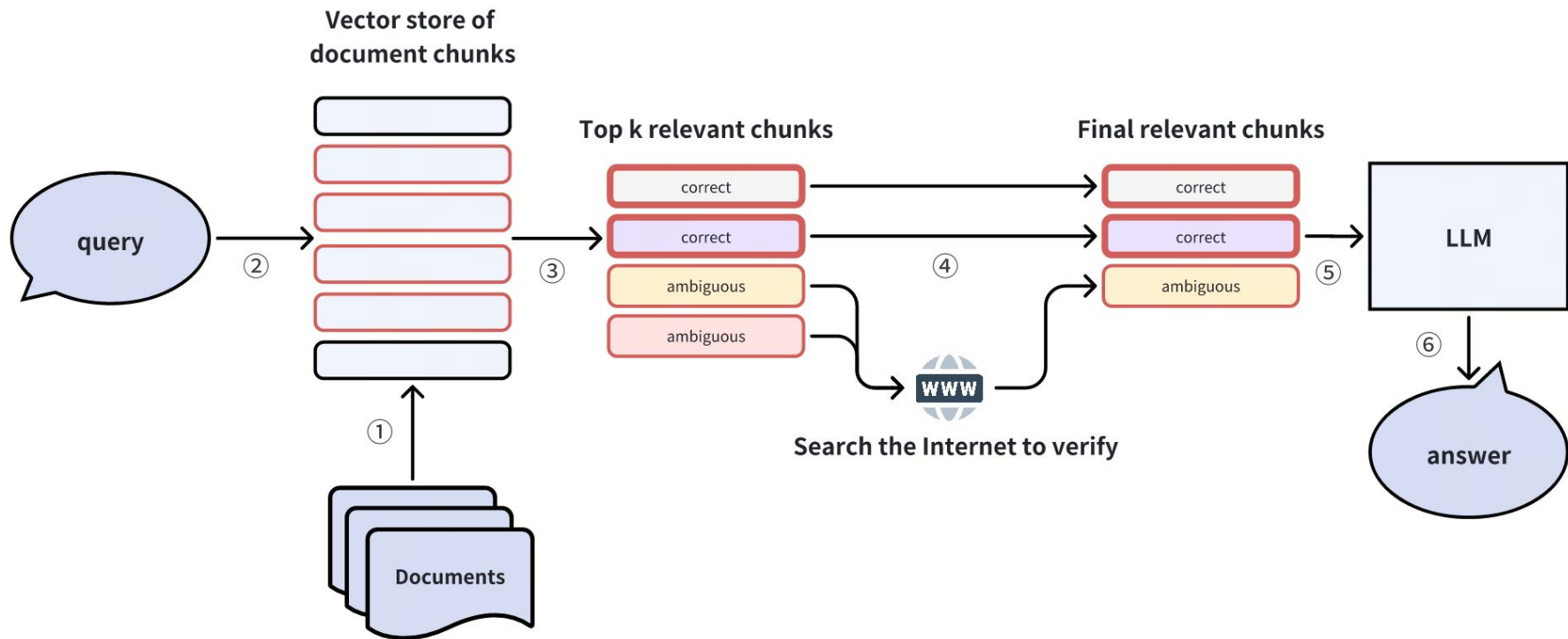


Agentic RAG

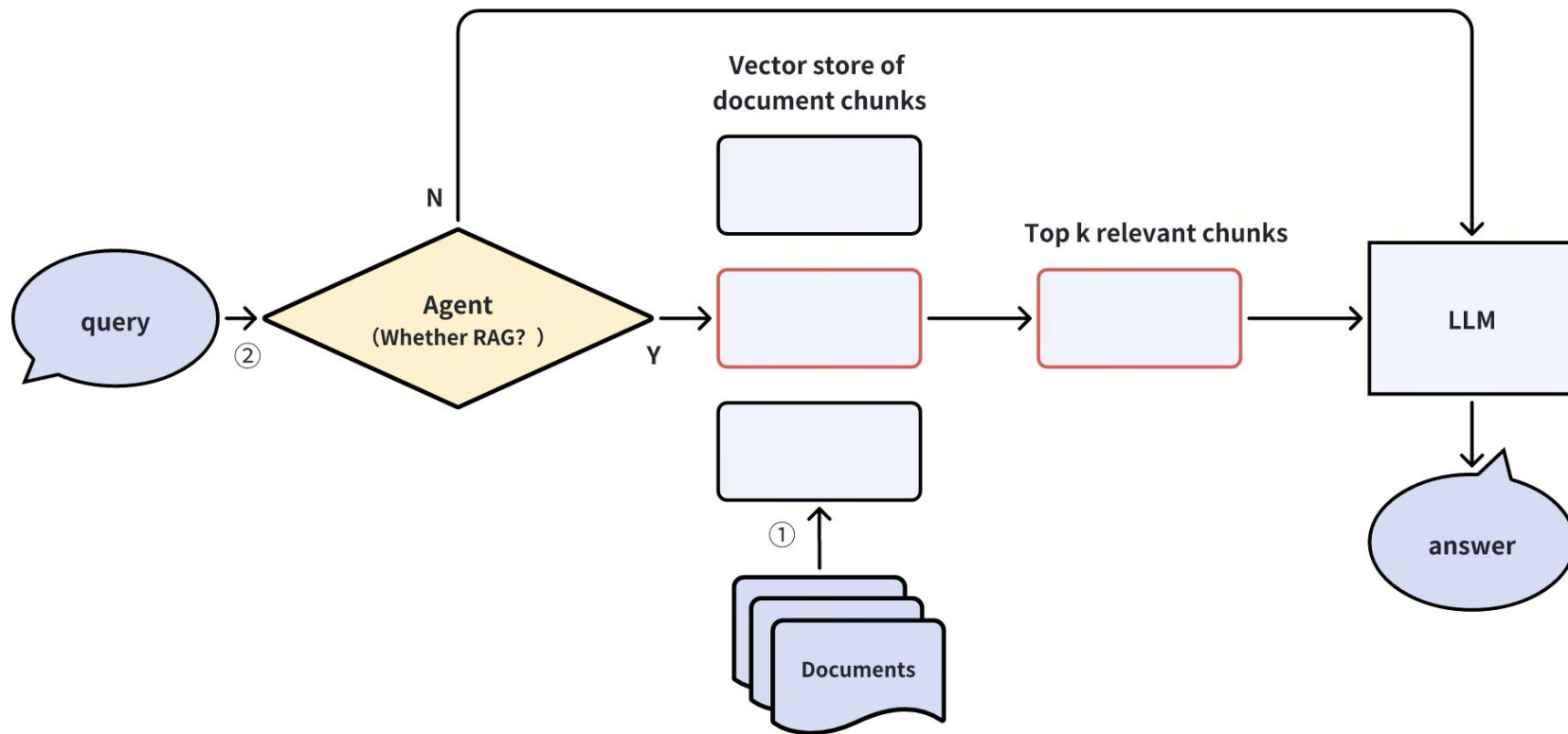


- ✓ Multi-turn
- ✓ Query / task planning layer
- ✓ Tool interface for external environment
- ✓ Reflection
- ✓ Memory for personalization

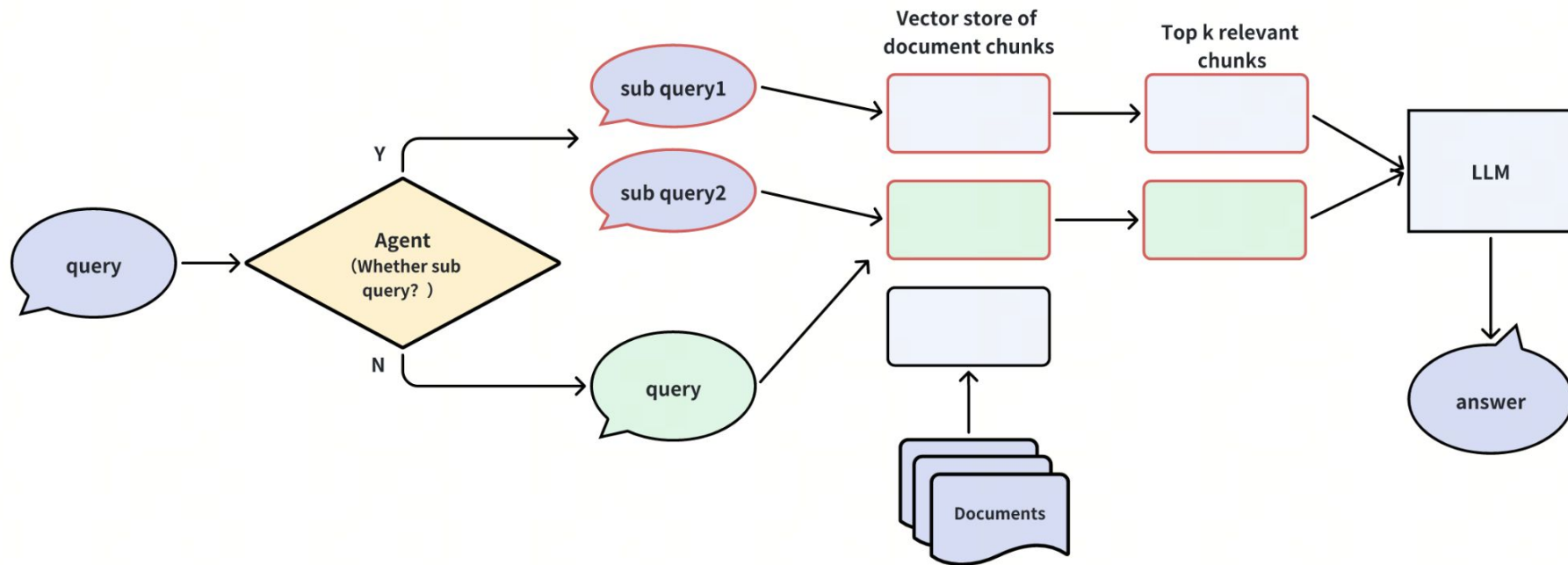
Self-reflection



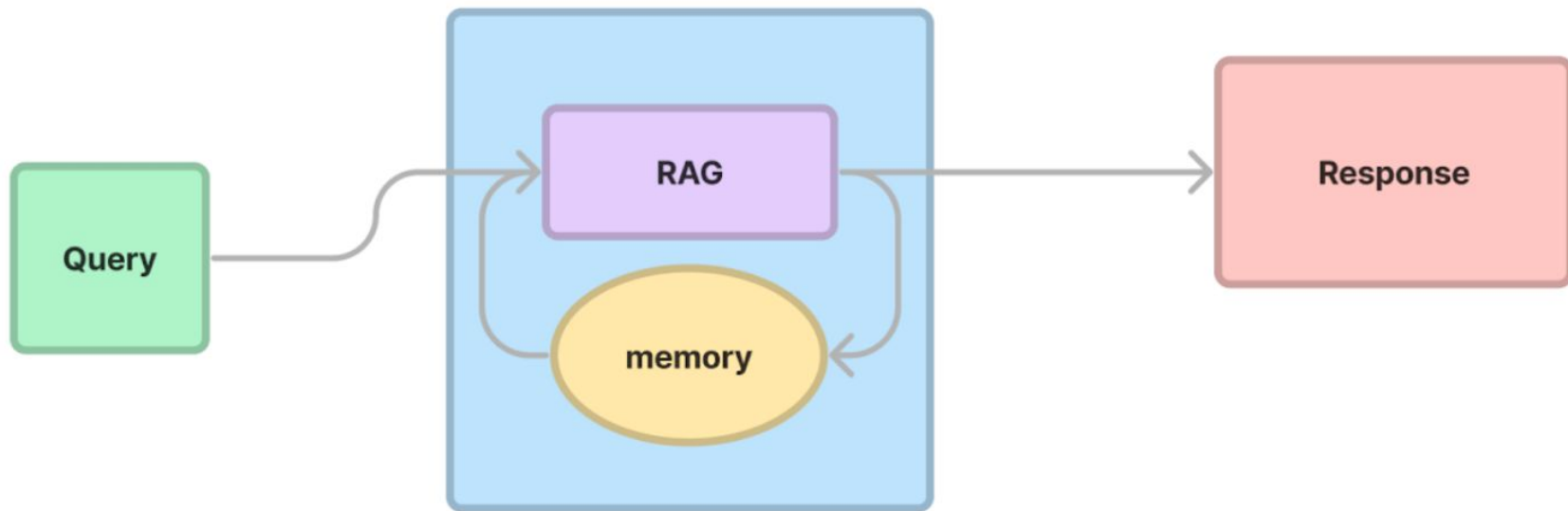
Query routing



Query routing with sub query

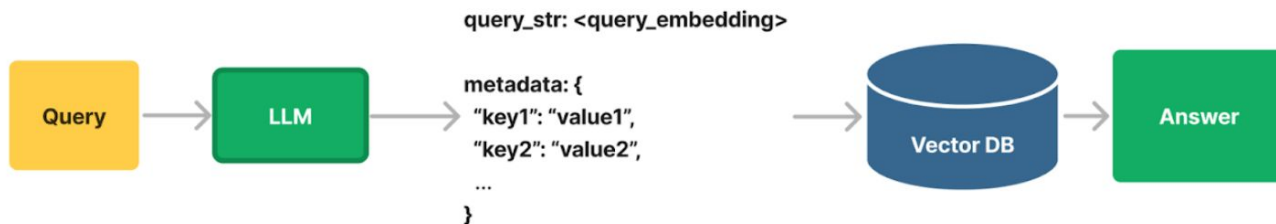


Conversation Memory

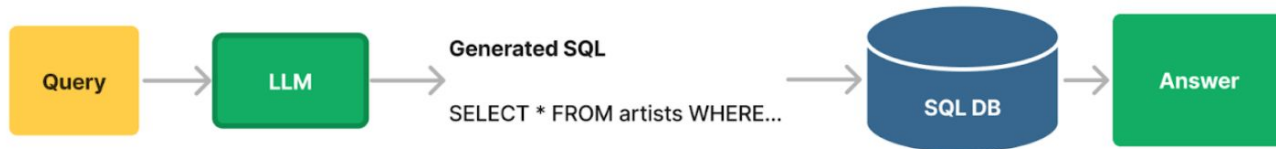


Tool Use

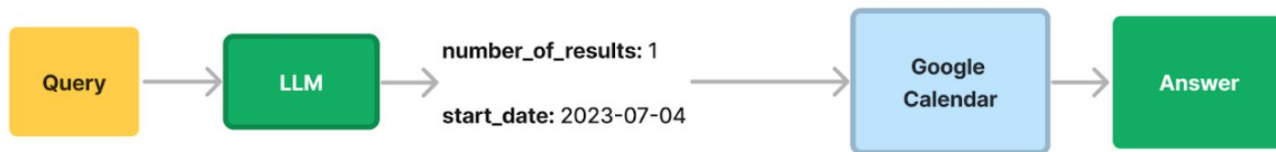
Auto-Retrieval



Text-to-SQL



Calendar

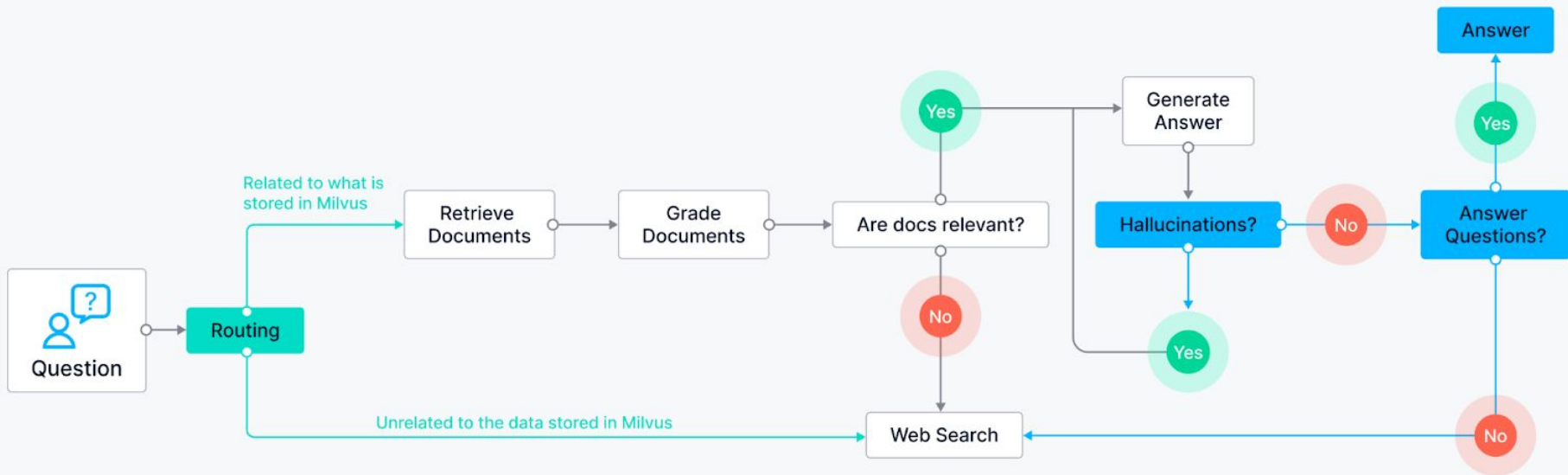


09

RAG in action with Milvus Lite



General Ideas





Demo!

Thank you



milvus.io



github.com/milvus-io/



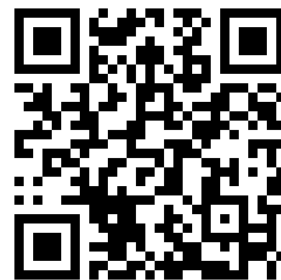
[@milvusio](https://twitter.com/milvusio)



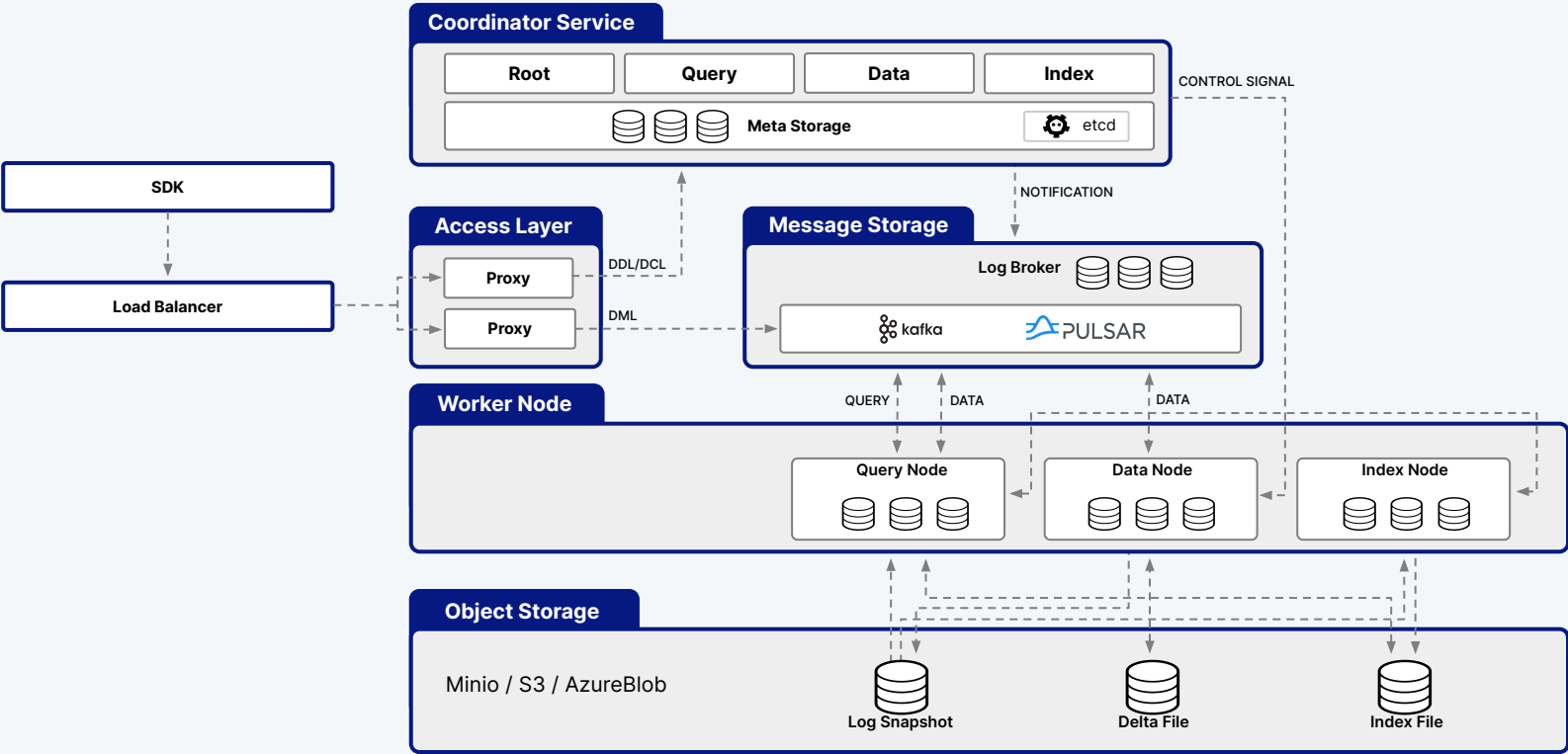
[@stephenbtl](https://twitter.com/stephenbtl)



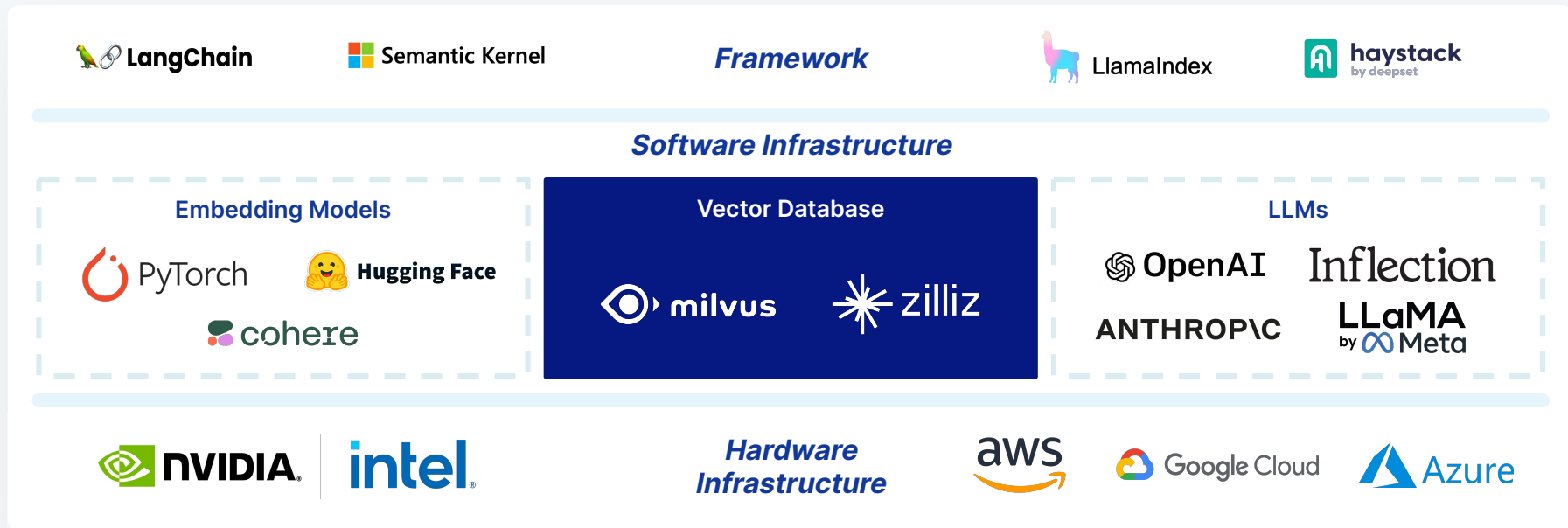
[in/stephen-batifol](https://www.linkedin.com/company/stephen-batifol/)



Milvus Architecture



Well-connected in LLM infrastructure to enable RAG use cases



Driving AI Innovation with leading business



COMPASS



Inflection

intuit.



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